

Python async TB

Cocotb tests are Python coroutines

async, await, and the decorator

- Three pieces fit together
- Async def declares a coroutine, the function can pause at await and come back later
- Await hands control to the event loop until a trigger fires
- The cocotb test decorator registers your coroutine so the runner actually starts it
- A plain def test cannot await

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Python async TB sketch

Sketch a cocotb-style **async def** test and **await** timeline: the coroutine suspends until a trigger resumes it. Starter: `async def test+await Timer(10) — DONE.`

Starter example: `@cocotb.test()+async def test+await Timer(10) — Run reaches DONE.`

Load starter example

Challenges 0 / 22 cleared

Quiz: **async def**: async def in a cocotb test means...

☐ the function is a coroutine that can suspend at await

☐ the DUT is synthesized automatically

☐ plusargs are disabled

☐ only Makefile PHONY targets run

Show hint

Check

Next

Idle

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Core ideas

async def

Coroutine that can suspend at await.

await

Yield until a trigger resumes you.

@cocotb.test

Registers the TB entry for the runner.

timeline

enter → suspend → resume → DONE.

Browser lab starter

learn_cocotb — python async tb

Real cocotb track practice

- In the real cocotb track, open this module's examples prompts
- Restate the async testbench idea in one sentence, coroutine plus await plus registration
- Sketch a tiny async test on paper
- Optional stretch
- No simulator required yet; the goal is to recognize the shape when you see it in code

Pitfalls to watch

- Do not use a plain def when you need await
- Do not forget the test decorator
- Do not assume await is instant
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load starter, run to DONE
- On paper, write one valid cocotb test skeleton with all three pieces named
- When you are ready, take the short quiz, then continue to cocotb triggers